

LESSON 13.1

ANGLE RELATIONSHIPS



LESSON INTRODUCTION

MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.

LEARNING TARGETS

- I can identify adjacent, supplementary, complementary, and vertical angles.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

ESSENTIAL QUESTIONS

- What are adjacent angles?
- How do I determine if two angles are adjacent angles?
- What are supplementary angles?
- What are complementary angles?
- How do I determine if angles are supplementary or complementary?
- What are vertical angles?
- How do I determine if two angles are vertical angles?

LESSON NARRATIVE

In this lesson, students explore angle relationships as part of defining and identifying adjacent, supplementary, complementary, and vertical angles. Foundational vocabulary is introduced in this lesson that will be used throughout the unit and in High School Geometry.

COMPONENT SUMMARY

13.1.1 WARM-UP (~5 MIN.)

Determine the weight of sand needed for a given context

13.1.3 EXPLORATION (15 MIN)

Explore angle relationships

13.1.5 WRAP-UP (~5 MIN)

Reflect on the lesson using a K-W-L chart

13.1.2 GUIDED INSTRUCTION (10 MIN)

Define and identify adjacent angles

13.1.4 BRING IT TOGETHER (15 MIN)

Define and identify complementary, supplementary, and vertical angles

RESOURCES

GLOSSARY

Key Terms:

- Adjacent Angles
- Complementary Angles
- Supplementary Angles
- Vertical Angles

Additional Terms:

- N/A

MATERIALS

- Straightedge
- Protractor
- 3 rectangular pieces of paper (i.e. index cards) per student or pair of students
- (Optional) Chart paper and markers
- (Optional) Small circular objects, such as tokens or coins

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think that all adjacent angles share a relationship similar to complementary and supplementary in all cases.
- Students may not recognize the properties of a straight angle.

THINGS TO CONSIDER

- 13.1.3 Exploration students may need assistance or modeling with completing the steps listed in the component. Be careful to only aid with completing the steps, and not giving away the discoveries that students will make.
- Consider having extra paper for students who may want to start over on a new piece of paper if they make mistakes.

LITERACY AND LANGUAGE ROUTINES

Take Turns

13.1.3 Exploration provides students an opportunity to work with a partner to explore angle relationships. Students can take turns to complete each task within the activity. For example, to complete question 3, one student can read the steps to their partner. While their partner is completing the steps, the reader is watching to make sure the steps are being followed accurately and assisting when necessary. Then, once the steps are completed, the partners switch roles so that both partners have had the opportunity to hear the steps and follow the steps on their own rectangular piece of paper. An alternative would be to allow partners to take turns completing the steps on the same piece of paper.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR1.1 Participate in Effortful Learning

13.1.3 Exploration engages students to help and support each other when attempting a new method or approach. Students explore angle relationships using a hands-on, collaborative activity. The activity is structured for students' experience reflective discussions to help them stay engaged and maintain a positive mindset through the activity.

DIFFERENTIATE INSTRUCTION

Consider creating and displaying a visual representation in 13.1.4 Bring It Together, such as an anchor chart or foldable, to highlight the vocabulary terms. Students can reference the visual representations throughout the remainder of the unit.

13.1.1 WARM-UP: SANDY BEACHES

TEACHER MOVES

If time permits, consider leveraging a Think-Pair-Share protocol when reviewing 13.1.1 Warm-Up. Allow the students time to discuss their answers and strategies for solving the question. Also, allow students the time to revise their answers, if they need, after their partner discussions.



13.1.1 Warm-Up: Sandy Beaches

There are so many beaches around the world that most people can't tell whether they are natural or man-made. For example, the 7.5-acre Crystal Lagoon in Wesley Chapel, FL includes a man-made beach.



1. To create a man-made beach, sand needs to be brought in to construct it. One cubic foot of sand weighs around 100 pounds. One acre covers about 43,560 square feet.

What is the weight of the sand that needs to be brought in to cover a 1-acre area with 3 inches, or $\frac{1}{4}$ foot, of sand?

1,089,000 lbs

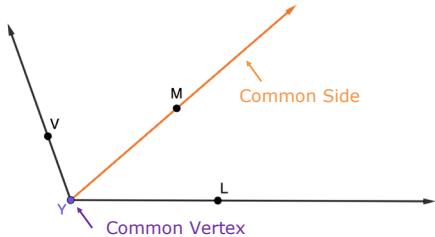


13.1.2 Guided Instruction: Adjacent Angles

Angles are formed when lines, line segments, or rays intersect.

Two angles that share a side and vertex and do not overlap are called **adjacent angles**.

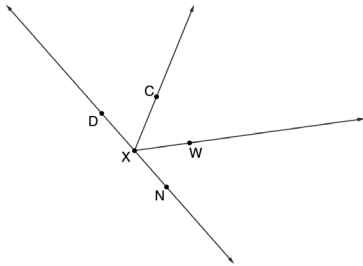
The figure shows an example of adjacent angles $\angle MYM$ and $\angle MYL$.



Point Y is the common vertex.
Ray $\overline{MY}$ is the common side that both angles share.

1. The measure of $\angle MYM$ is 68° and the measure of $\angle MYL$ is 41° .
- a. What is the measure of $\angle MYL$?
 109°
- b. Discuss with your partner how you know the measure of $\angle MYL$ from the definition of adjacent angles and the given information.
Answers will vary.

2. Line DN , ray XC , and ray XW intersect to form angles, as shown.



Complete the table to identify whether each pair of angles given are adjacent angles.

Angle Pair	Angle Relationship
$\angle CXD$ and $\angle CXW$	☐ adjacent angles ☐ not adjacent angles
$\angle CXW$ and $\angle DXW$	☐ adjacent angles ☐ not adjacent angles
$\angle DXW$ and $\angle WXN$	☐ adjacent angles ☐ not adjacent angles
$\angle NXC$ and $\angle DXC$	☐ adjacent angles ☐ not adjacent angles
$\angle DXC$ and $\angle NXW$	☐ adjacent angles ☐ not adjacent angles

13.1.2 GUIDED INSTRUCTION: ADJACENT ANGLES

DIFFERENTIATE INSTRUCTION

Consider having students create a Frayer Model for the terms in this unit. The Frayer Model includes the definition (from the book), characteristics, example, and non-example.

Students can benefit from a verbal think-aloud that connects the language with the naming of the angle to the common side and the common vertex.

TEACHER MOVES

Students decompose angles into non-overlapping angles and demonstrate their angle measure is additive in 4th grade (MA.4.GR.1.2 and MA.4.GR.1.3). The language of adjacent angles is new learning.

TEACHER MOVES

Consider leveraging a Poll the Class instructional routine protocol. Allow students a couple of minutes to determine the correct angle relationship. Once time is up, call out the angle pair and then poll the class to have them vote on the angle relationship choice. Have students share their rationale on why they selected that relationship. A class poll can also be created on digital platforms such as Kahoot, Microsoft Teams, etc.

13.1.3 EXPLORATION: ANGLE RELATIONSHIPS

TEACHER MOVES

Materials needed for this activity:

- 3 rectangular pieces of paper (i.e. index cards)
- Straightedge
- Protractor
- (Optional) Small circular objects, such as tokens or coins, for drawing semicircles

Students will need 3 papers each or 3 for each partner pair. The size of the papers doesn't matter, so they can be a half or quarter sheet of paper as long as they are cut using a straightedge, so all pieces have 4 right angles. Index cards could also be used.

ENRICHMENT

Can you determine the type of angle by only the image? Explain. If the square was not present in the first image, would you be able to state with certainty that the image is a right angle or that it measures 90 degrees exactly? Explain.

TEACHER MOVES

This exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection.

Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group 13.1.4 Bring It Together. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

TEACHER MOVES

Watch how students are using their tools, like the protractor, in this activity. Intervene if students are struggling with using the tools correctly.



13.1.3 Exploration: Angle Relationships

- Several angle names and measures are given in the bank below.

acute angle	obtuse angle	20°	45°	90°
right angle	straight angle	125°	180°	360°

Work with your partner to complete the table by matching the angle name and measure from the bank to each type of angle shown. Some words and angle measures from the bank will not be used.

Image	Angle Name	Angle Measure
	<i>Right Angle</i>	90°
	<i>Straight Angle</i>	180°

Your teacher will give you three rectangular pieces of paper to use for the remainder of the Exploration.

- Use one rectangular piece of paper to complete the following.
 - Draw a small half-circle in the middle of one side, as shown.



- Use a straightedge to draw a line starting from the center of the half-circle, all the way across the paper. The line can go in any direction or angle across the paper. Two possible examples are shown.



- Cut your paper along the line you drew.
- On each of the two pieces created, use a protractor to measure the angle that is marked by part of the circle. Then, label the angle measure on the paper.
- Write two observations about the two angles.
Answers will vary. The angle measures have a sum of 180°. One angle is larger, and one angle is smaller.

- f. Clare measured the angle on one of her pieces to be 70° . Discuss with your partner how to predict the angle measure of her other piece. Then, write your prediction.
I predict 110° because when the angles are adjacent and form a straight line their sum is 180° .

3. Use a second rectangular piece of paper to complete the following.

- a. Draw a small quarter-circle in one of the corners.



- b. Use a straightedge to draw a line starting from the vertex with the quarter-circle, all the way across the paper. The line can go in any direction or angle across the paper.
 c. Cut your paper along the line you drew.
 d. On each of the two pieces created, use a protractor to measure the angle that is marked by part of the circle. Then, label the angle measure on the paper.
 e. Add your two angle measures together. What do you notice?

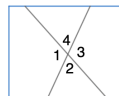
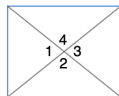
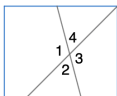
Answers will vary. The angle measures add to 90° .

- f. Priyanka measured the angle of one of her pieces to be 53° . Discuss with your partner how to predict the angle measure of her other piece. Then, write your prediction.

I predict 37° because the angles are adjacent and form a right angle which means their sum is 90° .

4. Use the third rectangular piece of paper to complete the following.

- a. Use a straightedge to draw two intersecting lines across the paper. Then, label the angles created with 1, 2, 3, and 4. Three possible examples are shown.



- b. Cut across the straight lines you drew, creating four separate pieces.
 c. Rotate and flip the angles to determine how they compare with each other. This could include laying angles on top of one another to compare their size. Summarize your observations of how the angles compare to each other.
Answers will vary.
 d. Compare your findings with your partner. Make a conjecture about the relationship between angle measures when two lines intersect.

Answers will vary. When two lines intersect, the angles opposite one another have equal measure. When two lines intersect, there are four angles but only two unique angle measures.

13.1.3 EXPLORATION: ANGLE RELATIONSHIPS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider having some of these pre-cuts for students who may struggle with completing the steps of cutting the paper correctly. If you choose to use this method, still ensure that students get the opportunity to complete questions 2d-2f.

TEACHER MOVES

Listen for student responses to question 3f. Redirect students who are making predictions that aren't related to this activity.

TEACHER MOVES

At the conclusion of the activity, leverage student responses to question 4d as a springboard for discussion with the entire class as it connects to 13.1.4 Bring It Together.

13.1.4 BRING IT TOGETHER: ANGLE RELATIONSHIPS

TEACHER MOVES

The definitions used here in this section come directly from the B.E.S.T. Glossary. Use the definitions provided to match the expectations for understanding of each of these terms.

ENRICHMENT

Do you think it's possible for angles to be both supplementary and complementary? Why or why not?

ENRICHMENT

Do you think that two adjacent angles will always be supplementary? Why or why not?

ENRICHMENT

Can two angles be complementary without being adjacent? Why or why not?



13.1.4 Bring it Together: Angle Relationships

Three different angle relationships were explored in the Exploration. Their definitions are shown.



Two angles whose measures add up to 90° are **complementary angles**.



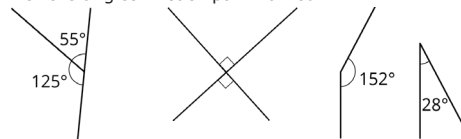
Two angles with measures that add up to exactly 180° are **supplementary angles**.



Vertical angles are the opposite angles formed when two lines intersect.

- Complete the statements using the definitions of each angle relationship.
 - The angles created in question 2 of the Exploration are supplementary angles.
 - The angles created in question 3 of the Exploration are complementary angles.
 - The angles created in question 4 of the Exploration are vertical angles.

- Supplementary angles can be adjacent or non-adjacent. Three examples of supplementary angles are shown, with the angles in each pair marked.



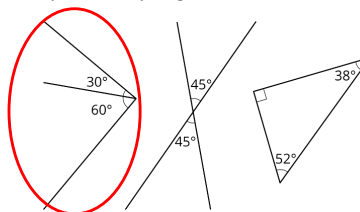
Complete the statement.

When supplementary angles are adjacent, they form

- ☐ an acute
- ☐ a right
- ☐ an obtuse
- ☒ a straight

angle together.

- Three examples of complementary angles are shown. Complementary angles can be adjacent or non-adjacent.
 - Circle the example that shows an adjacent pair of complementary angles.



- b. Complete the statement.
- When complementary angles are adjacent, they form

☐ an acute

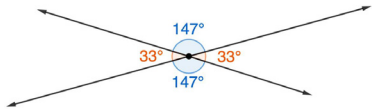
☒ a right

☐ an obtuse

☐ a straight

 angle together.

4. Each pair of vertical angles formed by the intersecting lines below are shown in a different color.



- a. Discuss with your partner the angle relationship between adjacent angles when two lines intersect. *Answers will vary. The adjacent angles formed are supplementary.*
- b. Complete the statements.
- Vertical angles

☐ sometimes

☒ always

 have equal measures. Pairs of adjacent angles created when two lines intersect are

☐ complementary.

☒ supplementary.

13.1.4 BRING IT TOGETHER: ANGLE RELATIONSHIPS (CONTINUED)

TEACHER MOVES

Consider leveraging a Notice and Wonder protocol for the image in question 4. Ask students to share what they notice about the angles and what they wonder about them, prior to releasing them to trying questions 4a and 4b on their own.

ENRICHMENT

Why do you think vertical angles will always have equal measures?

 13.1.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the lesson that I knew were ...	Something I wondered during the lesson was ...	Something I learned during the lesson was ...
<i>Answers will vary.</i>	<i>Answers will vary.</i>	<i>Answers will vary.</i>

13.1.5 WRAP-UP: REFLECTION

TEACHER MOVES

Consider pairing students to discuss their reflections to see if their peers wondered something similar and/or learned something similar.